Kalasalingam Academy of Research and Education

ResumeX AI – AI-Powered Resume Analysis and Career Guidance Platform

Project ID: PRJ013

Team ID: Gladiators

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1. Introduction

In today's digital environment, Generative Artificial Intelligence (GenAI) is transforming the way individuals interact with technology and access intelligent solutions. However, students and job seekers often face difficulties in preparing professional resumes, understanding job requirements, identifying skill gaps, and improving their career profiles.

ResumeX AI is a Generative AI-powered resume analysis and career guidance platform developed to help users evaluate their resumes and improve their job readiness. The application uses Google Gemini through API integration to support intelligent resume analysis, personalized recommendations, and AI-generated career guidance. The system analyzes uploaded resumes against provided job descriptions to identify matching skills, missing skills, partially matching skills, and additional skills. It also provides career goal and role-readiness analysis to help users understand their suitability for specific job opportunities.

The application supports multiple intelligent features, including resume section rewriting, personalized career learning roadmaps, interview question generation, and multi-job comparison. Users can also access their previous analyses, bookmark results, and download analysis reports in PDF format. The primary goal of ResumeX AI is to simplify resume evaluation and career preparation by providing an intelligent, user-friendly, and Generative AI-powered platform that helps candidates improve their professional profiles and prepare for employment opportunities.

2 Problem Statement & Objectives

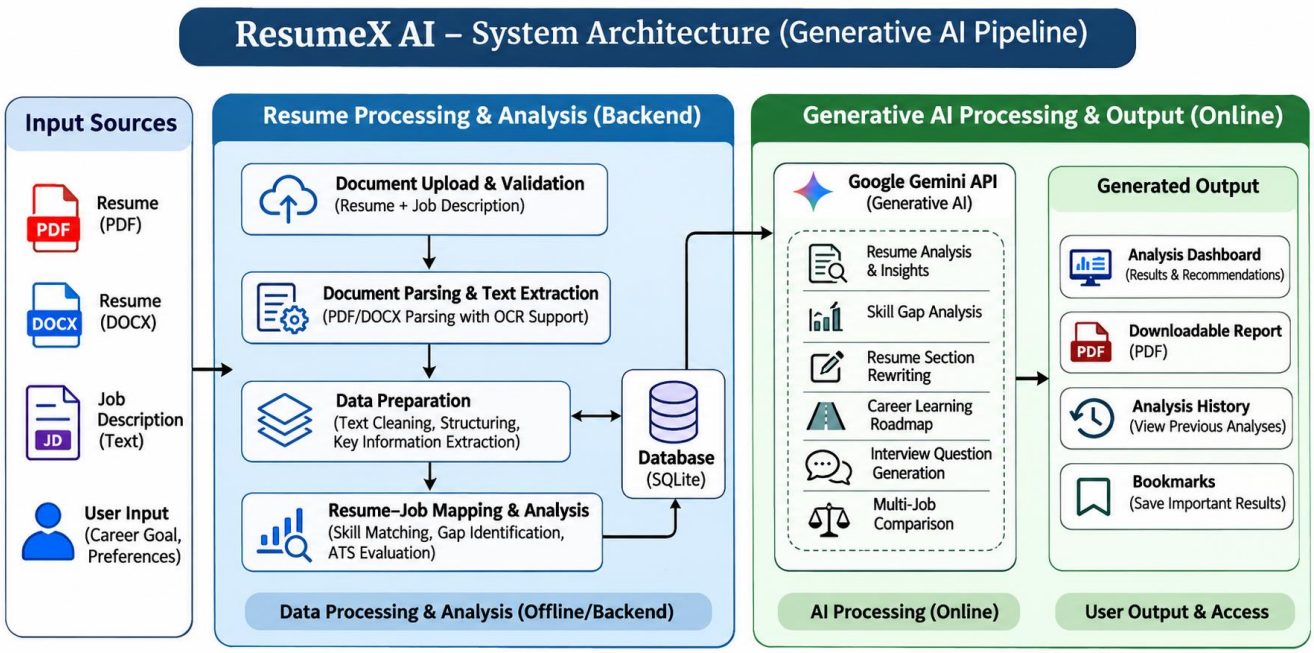
Problem: Students and job seekers often face difficulties in preparing effective resumes that meet specific job requirements. Traditional resume evaluation methods mainly rely on keyword matching and predefined criteria, which may fail to identify the contextual relevance of skills and qualifications. Candidates also struggle to identify missing skills, improve resume content, understand their job readiness, and prepare for interviews. Manual resume evaluation and career preparation can be time-

consuming and require external guidance.

Objectives:

- To develop a Generative AI-powered platform for analyzing resumes against specific job descriptions.
- To integrate Google Gemini API for intelligent resume analysis and AI-generated recommendations.
- To identify matching, missing, partially matching, and additional skills based on job requirements.
- To provide AI-powered resume section rewriting and content improvement suggestions.
- To generate personalized career learning roadmaps based on identified skill gaps.
- To generate relevant interview questions to support candidate preparation.
- To implement multi-job comparison to evaluate resume suitability across different job opportunities.
- To help students and job seekers understand their career readiness and improve their employment preparation.

3 System Architecture



3.1 Input and Document Processing phase

Users upload their resumes in PDF or DOCX format and provide job descriptions. The Flask backend extracts resume content using document parsing libraries and OCR support. The extracted information is processed and prepared for further analysis.

3.2 Resume Analysis and Generative AI Processing Phase

The processed resume information and job requirements are analyzed, and Google Gemini API supports AI-powered analysis and content generation. The system provides skill gap analysis, resume improvement suggestions, career learning roadmaps, and interview questions.

3.3 Output and Report Generation Phase

The generated results are displayed through the React frontend. Users can view analysis results, compare multiple job opportunities, access previous analyses, bookmark important results, and download PDF reports.

4 Implementation Details & Tech Stack

ResumeX AI is implemented as a web-based application that combines Generative AI capabilities with resume processing and career guidance functionalities. The frontend provides an interactive interface, while the Python Flask backend manages document processing, application logic, database operations, and communication with the Google Gemini API.

Major Modules

1. User Authentication: Provides secure user registration and login using password hashing and JWT-based authentication.

2. Resume Processing: Accepts PDF and DOCX resumes and extracts their text using document-processing libraries and OCR support.

3. AI Resume Analysis: Uses Google Gemini API integration to support resume analysis against job descriptions and generate relevant insights.

4. Skill Gap Analysis: Identifies matching, missing, partially matching, and additional skills to help candidates understand their qualifications.

5. Multi-Job Comparison: Allows users to compare their resumes against multiple job descriptions.

6. Resume Improvement: Provides AI-assisted rewriting suggestions for improving

resume sections.

7. Career Guidance: Generates career learning roadmaps and interview questions to support job preparation.

8. Database Management: Uses SQLite to maintain user information, analysis records, multi-job comparisons, and activity logs.

9. PDF Report Generation: Uses ReportLab to generate downloadable reports containing analysis results.

Technology Used	Purpose
Python	Backend development and application logic
React 18	Frontend user interface
JavaScript	Frontend functionality
Vite	Frontend development and build tool
Tailwind CSS	User interface styling
Flask	Backend framework and API development
Google Gemini API	Generative AI-based analysis and content generation
SQLite	Database management
pypdf	PDF text extraction
python-docx	DOCX document processing
Tesseract OCR	Text extraction from scanned documents
ReportLab	PDF report generation
JWT	User authentication
Pytest	Backend testing

5 Results and Testing

5.1 Results

The ResumeX AI application provides an interactive platform for analyzing resumes and generating personalized career recommendations using Generative Artificial Intelligence.

The application successfully integrates resume document processing, job description analysis, Google Gemini API integration, database management, and PDF report generation.

The major results and functionalities include:

1. Successful uploading and processing of PDF and DOCX resumes.
2. Extraction of resume content using document parsing and OCR support.
3. AI-powered resume analysis against job descriptions.
4. Identification of matching, missing, partially matching, and additional skills.
5. Generation of personalized career goal and role-readiness insights.
6. AI-assisted resume section rewriting and improvement suggestions.
7. Generation of personalized career learning roadmaps.
8. Generation of relevant interview questions.
9. Comparison of resumes against multiple job opportunities.
10. Storage and retrieval of previous analyses and bookmarked results.
11. Generation and downloading of PDF analysis reports.

5.2 Testing

The ResumeX AI application is evaluated through functional testing to verify the correctness and reliability of its major features.

Testing covers user authentication, resume uploading, document parsing, job description analysis, AI integration, skill gap identification, multi-job comparison, resume rewriting, career roadmap generation, interview question generation, and PDF report generation.

The project README reports 17 automated tests covering AI functionality, API operations, and authentication/database operations.

- AI functionality: 6 tests.
- API functionality: 6 tests.
- Authentication and database: 5 tests.

5.3 Performance Evaluation

The application's performance can be evaluated using the following parameters:

- Resume document processing time.
- Resume and job description analysis response time.
- Relevance of identified skills.
- Quality of AI-generated recommendations.
- Accuracy and usefulness of resume improvement suggestions.
- Response generation time.
- Reliability of database operations.
- PDF report generation time.

6 Conclusion & Future Scope

Conclusion: ResumeX AI is a Generative AI-powered resume analysis and career guidance platform developed to simplify the resume evaluation and job preparation process. The application integrates Google Gemini API to provide intelligent resume analysis, skill gap identification, personalized career recommendations, and AI-generated content. It combines resume document processing, job description analysis, AI integration, and database management within a single web application.

The platform provides several useful functionalities, including multi-job comparison, resume section rewriting, personalized career learning roadmaps, and interview question generation. These features help students and job seekers understand their professional strengths, identify areas for improvement, and prepare for employment opportunities. Overall, ResumeX AI demonstrates how Generative Artificial Intelligence can be applied to career development by providing an interactive, intelligent, and user-friendly solution.

Future Scope: In the future, ResumeX AI can be enhanced by implementing an advanced Retrieval-Augmented Generation (RAG) architecture using document embeddings and vector databases to improve context-based information retrieval. Integration with multiple Large Language Models can provide greater flexibility in AI-powered analysis and content generation. The platform can also be extended with real-time job recommendations through job portal APIs, LinkedIn profile integration, and multilingual support to make career guidance more accessible.

Additionally, advanced resume evaluation techniques can be introduced to provide more detailed insights into candidate suitability for specific job roles. Cloud-based infrastructure and improved scalability can support a larger number of users. Further enhancements in personalized learning recommendations and career guidance can make ResumeX AI a more

comprehensive Generative AI-powered career development platform.